BandUp Database: The Complete Guide

Every table and every idea, explained with one example world: the same people, the same exercises and the same answers from the first page to the last.

PostgreSQL 16 · 28 tables · migrations V1–V4 · Spec: `.kiro/specs/database/` · Diagram: `schema.dbml` (dbdiagram.io)

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1 Database words you need

A database is a set of **tables**, like Excel sheets. Each **row** is one thing (one user, one question), and each **column** is one property (name, email). These words appear on every page of this guide:

Word	Meaning	Example
Primary key (PK)	The column that identifies a row. No two rows share it.	`app_user.id = 2` is always Aysel
Foreign key (FK)	A column that points to a row in another table. The database refuses to point at something that doesn't exist.	`user_role.user_id = 2` → Aysel. `user_id = 999` is rejected.
Unique	No duplicates allowed in this column (or combination).	Two exercises can't both have slug `yellowstone-wolves`
NULL	"Empty / not set". **NOT NULL** means the value is required.	`duration_seconds = NULL` means no timer
CHECK	A rule the value must follow.	`duration_seconds > 0`, so −5 is rejected
Index	A lookup shortcut that makes searches fast, like a book's index.	Find all of Aysel's attempts instantly
Cascade / Restrict	What happens to child rows when a parent is deleted: **cascade** deletes them too, **restrict** refuses the delete.	Delete a question → its options go too (cascade). Delete an exercise with results → refused (restrict).
slug	A readable text id used in URLs.	`/reading/yellowstone-wolves`
uuid	A long random id that can't be guessed.	`7f3c9a10-…-e2`
jsonb	A column that stores a small JSON value.	`"FALSE"` or `["B","A"]`
timestamptz	Date and time with time zone (stored in UTC).	`2026-10-22 09:14 UTC`
Migration	A numbered SQL file that changes the database. They run in order, once each.	`V1__core_schema.sql`, `V2__seed.sql`

2 The example world

Four people use BandUp. They appear in every example:

Murad
Admin and author. Creates exercises. Speaks Azerbaijani.

Aysel
Learner from Baku. Help text in **Azerbaijani**.

Emre
Learner from Istanbul. Help text in **Turkish**.

Leyla

			Translator. Adds Turkish translations. Can't change questions.

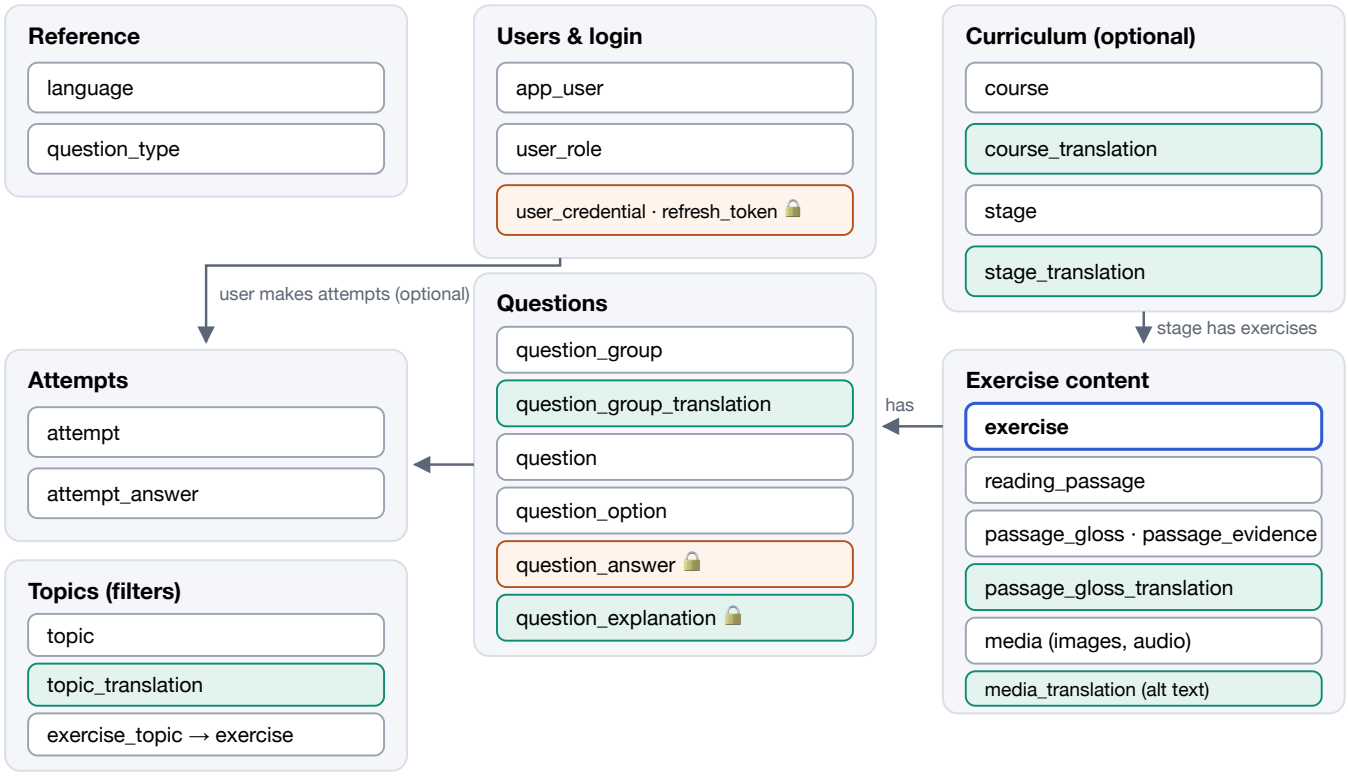
And five exercises:

Exercise	What it is	State
The Return of the Wolves	Mini reading: short passage, 7 questions (8 points), 5-minute timer	Published, learners use it
Coral Reefs	Full IELTS-size passage with matching headings and a diagram	Draft, Murad is still writing it
Present Perfect 1	Grammar drill, no timer, no passage	Draft, planned for the future
Busy Honeybees	Mini reading, A2, **standalone**: not in any course	Published
Ancient Rome	Full IELTS-size passage, C1, standalone	Published

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3 The map: 28 tables in 7 groups

Green tables hold translations (one row per language). 🔒 tables are never shown to learners before they submit.



All green translation tables and every language column reference language(code).
🔒 = never read by the learner endpoint before Submit. Courses/stages are optional and never lock anything.

The migrations so far (they run in order, once, and are never edited)

File	What it does
`V1__core_schema.sql`	creates the 26 content, learner and reference tables
`V2__seed_reference_and_demo.sql`	languages, 13 question types, 5 topics, the IELTS Reading course, the Wolves exercise (Azerbaijani glosses)
`V3__identity_login_tables.sql`	login tables: `user_credential`, `refresh_token`
`V4__english_gloss_definitions.sql`	simple English definitions for the 25 Wolves words (the fallback safety net)

4 The full story, step by step

This is how the tables fill up in real life. Each step names the tables it touches. The following sections show each table in detail.

1. **Setup (once).** The first migration creates the languages (English, Azerbaijani, Turkish…) and the 13 question types.
   → `language`, `question_type`

2. **Murad gets an account** with the ADMIN and AUTHOR roles and a password (stored only as a hash).
   → `app_user`, `user_role`, `user_credential`

3. **(Optional) Murad creates a course** "IELTS Reading" with Stage 1 "Beginner", and titles in English and Azerbaijani. Exercises don't need one: Honeybees and Ancient Rome have no course.
   → `course`, `course_translation`, `stage`, `stage_translation`

4. **Murad creates the Wolves exercise**: title, 5-minute timer, format **MINI**, level **B1**, topic **Nature**, and the passage. He marks difficult words (glosses) and the sentences that prove each answer (evidence).
   → `exercise`, `exercise_topic`, `reading_passage`, `passage_gloss`, `passage_gloss_translation`, `passage_evidence`

5. **Murad adds 3 question groups**: True/False/Not Given, sentence completion and "Choose TWO", with answers and explanations.
   → `question_group`, `question`, `question_option`, `question_answer`, `question_explanation`

6. **Murad publishes it.** Status changes from DRAFT to PUBLISHED, and learners can now see it.

7. **Aysel logs in.** Her password is checked against the stored hash. She gets a short access token and a refresh token; only the refresh token's hash is saved.
   → `user_credential`, `refresh_token`

8. **Aysel filters** "Reading → Mini → Nature", picks Wolves, practises and submits. The server grades her answers and saves the result.
   → `attempt`, `attempt_answer`

9. **Murad adds an extra accepted answer** ("river banks" for Q4). The exercise revision goes from 1 to 2. Aysel's old result doesn't change.

10. **Leyla translates** some word meanings into Turkish and reviews the English definitions, which were drafted by a machine.
    → `passage_gloss_translation`, `question_explanation`

11. **Emre practises in Turkish.** Translated words show in Turkish. Missing ones fall back to English.

5 Group 1: Reference tables

Small lists that rarely change. Other tables point to them.

`language`

Reference

Every language BandUp knows. Every other table that stores something "in a language" points here.

code (PK)	name_en	name_native	direction	fallback_code	is_target	is_support	enabled
en	English	English	ltr	—	true	true	true
az	Azerbaijani	Azərbaycanca	ltr	en	false	true	true
tr	Turkish	Türkçe	ltr	en	false	true	true
ru	Russian	Русский	ltr	en	false	true	false
ar	Arabic	العربية	rtl	en	false	true	false

Good to know

- `code` is a standard language code (BCP-47): `en`, `az`, `pt-BR`.
- `direction`: `rtl` means right-to-left (Arabic, Persian), so the page layout must flip.
- `fallback_code`: which language to use when a translation is missing (section 11).
- `is_target`: can be *learned* (English). `is_support`: can be a learner's *help* language.
- `enabled = false`: the language exists but isn't offered yet. Russian is prepared but switched off.

`question_type`

Reference

The list of question types. The `family` decides how a type is answered and graded.

code (PK)	family	code (PK)	family
TFNG	choiceFixed	MATCH_HEADINGS	match
YNNG	choiceFixed	MATCH_INFO	match
MCQ	mcSingle	MATCH_FEATURES	match
MCMA	mcMulti	MATCH_ENDINGS	match
FILL	gap	SUMMARY_BOX	match
SUMMARY	gap	DIAGRAM_LABEL	gap
SHORT	gap		

Good to know

- Adding a new type in the same family is just a new row. The grading code works per family, not per type.
- All types are explained in `ai-docs/IELTS-Reading-Question-Types.pdf`.

6 Group 2: Users

`app_user`

Users

One row per person with an account. (Named `app_user` because `user` is a reserved word in PostgreSQL.)

id	public_id	email	display_name	ui_language	support_language	target_language	status
1	a1b2…	murad@example.com	Murad	az	az	en	ACTIVE
2	c3d4…	aysel@example.com	Aysel	az	az	en	ACTIVE
3	e5f6…	emre@example.com	Emre	tr	tr	en	ACTIVE
4	g7h8…	leyla@example.com	Leyla	en	tr	en	ACTIVE

Good to know

- Three languages per person: **ui_language** for buttons and menus, **support_language** for help such as word meanings and explanations, and **target_language** for what they are learning. Leyla reads the site in English but works on Turkish.
- `id` is internal. `public_id` (uuid) is what the API shows, so nobody can guess "user 3".
- Passwords are not here. The future login spec adds its own table.

`user_role`

Users

What each person is allowed to do. One person can have several roles.

user_id	role
1 (Murad)	ADMIN
1 (Murad)	AUTHOR
2 (Aysel)	LEARNER
3 (Emre)	LEARNER
4 (Leyla)	TRANSLATOR

Role	Can
LEARNER	practise, see own results
AUTHOR	create and edit exercises
TRANSLATOR	edit translations only
ADMIN	everything, incl. users

`user_credential` 🔒

Users · login

A user's password, stored **only as a BCrypt hash**. One row per user; it's deleted with the user.

user_id (PK)	password_hash	updated_at
1 (Murad)	$2a$10$N9qo8uLOickgx2ZMRZoMye…	2026-10-22 08:00
2 (Aysel)	$2a$10$7EqJtq98hPqEX7fNZaFWoO…	2026-10-22 08:55

Good to know

- A hash can't be turned back into the password. At login, the typed password is hashed the same way and the two hashes are compared.
- It's a separate table, not a column on `app_user`, so Google sign-in can be added later as its own table without touching this one.

`refresh_token` 🔒

Users · login

One row per refresh token handed out. The browser keeps the real token in a secure cookie; the database stores only its SHA-256 hash.

id	user_id	token_hash	family_id	issued_at	expires_at	replaced_by	revoked_at
1	2 (Aysel)	9f86d0…	f1	Oct 22 09:00	Nov 5 09:00	2	—
2	2 (Aysel)	3c2b1a…	f1	Oct 22 09:30	Nov 5 09:30	—	—
3	3 (Emre)	a71e4f…	f2	Oct 22 11:00	Nov 5 11:00	—	Oct 22 12:10

How to read it

- **Rotation:** every refresh gives Aysel a new token (row 2) and links the old one to it (`replaced_by = 2`). Both belong to one login session, family `f1`.
- **Theft protection:** if row 1's old token is ever used again, someone copied it, so the whole family `f1` is revoked and Aysel must log in again.
- **Logout:** Emre logged out, so his token has `revoked_at` set and stops working.
- The database refuses a hash that isn't 64 lower-case hex characters, a duplicate hash, and an expiry time before the issue time.

7 Group 3: Courses, stages and topics

> **All of this is optional. Nothing is ever locked.**
> An admin can publish a mini reading on its own, and any learner can open and submit any published exercise. Courses and stages are only an optional way to group exercises (like a playlist). Topics are labels for filtering.

Courses and stages group exercises, like a textbook with chapters:

```
📘 IELTS Reading                    ← course
   ├── Stage 1 · Beginner            ← stage
   │      └── #4 The Return of the Wolves   ← exercise (position 4)
   └── Stage 2 · Intermediate
          └── #1 Coral Reefs
📗 Grammar Basics (draft)
Present Perfect 1, Busy Honeybees, Ancient Rome   ← standalone exercises (no course)
```

`course` — Curriculum

A whole subject, such as "IELTS Reading" or "Grammar Basics".

id	slug	skill	target_language	position	status
1	ielts-reading	READING	en	1	PUBLISHED
2	ielts-listening	LISTENING	en	2	DRAFT
3	grammar-basics	GRAMMAR	en	3	DRAFT

`position` = the order in the main menu. There's no title column, because titles are translated (next table).

`course_translation` — Curriculum · translation

The course title and description in each language.

course_id	language_code	title	description
1	en	IELTS Reading	Practise every IELTS reading question type.
1	az	IELTS Oxu	IELTS oxu sual növlərinin hamısını məşq edin.
1	tr	IELTS Okuma	—

Key = (course_id, language_code), so there's only one Azerbaijani title per course.

`stage` — Curriculum

A level or chapter inside a course.

id	course_id	position
1	1	1
2	1	2

`stage_translation` — translation

Stage titles per language.

stage_id	lang	title
1	en	Stage 1 · Beginner
1	az	Mərhələ 1 · Başlanğıc
2	en	Stage 2 · Intermediate

For an exercise in a stage, the reading page label **"Mərhələ 1 · Başlanğıc · Mini passage 4"** is built from the stage title plus the exercise's position. A standalone exercise shows just its format, such as "Mini test".

Topics: labels for filtering

`topic` — Topics

A subject label: Nature, Science, History…

`topic_translation` — translation

The topic name in each language.

id	slug	position
1	nature	1
2	science	2
3	history	3

topic_id	lang	name
1	en	Nature
1	az	Təbiət
1	tr	Doğa
3	az	Tarix

exercise_topic

Topics · link table

Which exercise has which topics. One exercise can have several (Honeybees = Nature + Science).

exercise_id	topic_id
1 (Wolves)	1 (nature)
2 (Coral Reefs)	1 (nature)
2 (Coral Reefs)	2 (science)
4 (Honeybees)	1 (nature)
4 (Honeybees)	2 (science)
5 (Ancient Rome)	3 (history)

A table like this, with only two ids, is called a **link table**. It's how "many to many" works: many exercises ↔ many topics.

8 Group 4: Exercise content

`exercise`

Exercise · the center of everything

One practice unit of any skill and any size. Almost every other table connects to it.

id	slug	skill	format	level	stage_id	pos	title	duration_seconds	status	revision
1	yellowstone-wolves	READING	MINI	B1	1	4	The Return of the Wolves	300	PUBLISHED	2
2	coral-reefs	READING	FULL	B2	2	1	Coral Reefs in Danger	1200	DRAFT	1
3	present-perfect-1	GRAMMAR	MINI	A2	—	—	Present Perfect 1	*null*	DRAFT	1
4	busy-honeybees	READING	MINI	A2	—	—	Busy Honeybees	240	PUBLISHED	1
5	ancient-rome	READING	FULL	C1	—	—	Ancient Rome	1200	PUBLISHED	1

`slug`	The URL id: `/reading/yellowstone-wolves`.
`skill`	READING, LISTENING, GRAMMAR… One table serves all skills.
`format`	**MINI** (short practice) or **FULL** (IELTS-size passage). The admin chooses; learners filter by it.
`level`	Difficulty A1–C2 (CEFR). Optional. Learners filter by it.
`target_language`	The language being tested (`en`). Not shown above; `en` for all of them.
`stage_id, position`	Optional place in a course. Empty for standalone exercises (Honeybees, Ancient Rome).
`duration_seconds`	Timer. 300 = 5 minutes. *null* = no timer (grammar drill).
`status`	DRAFT / PUBLISHED / ARCHIVED (section 15).
`low_time_threshold_seconds`	When the timer turns red. Wolves: 60 = the last minute (not shown above).
`revision, version`	Two different counters (section 14). `version` not shown above.
`created_by, updated_by, …_at`	Who changed it and when (Murad, 2026-10-22).

`reading_passage`

Exercise · READING only

The passage text for reading exercises. Exactly one row per reading exercise (`exercise_id` is its PK).

exercise_id	body_html	word_count
1	`<span class="ev" id="e1">By the 1920s, grey wolves had been completely <span class="w" data-g="g1">eliminated</span> from Yellowstone … largely through <span class="w" data-g="g2">government-sponsored</span> hunting</span>. Without their main <span class="w" data-g="g3">predator</span>, …`	182

Two markers live inside the HTML:

- `<span class="w" data-g="g1">` = a difficult word with gloss key **g1** (its meaning is stored in the gloss tables).
- `<span class="ev" id="e1">` = an evidence sentence with key **e1** (used by "Show in text").
- Listening will get its own table (`listening_audio`) next to this one. The grammar drill needs none.

`passage_gloss`

Exercise

Each marked word in a passage.

`passage_evidence`

Exercise

Each evidence sentence in a passage.

id	ex.	key	surface	lemma
11	1	g1	eliminated	eliminate
12	1	g2	government-sponsored	—
13	1	g3	predator	predator
14	1	g4	elk	elk

`surface` = the word as written; `lemma` = its dictionary form (for future word lists).

id	exercise_id	evidence_key
21	1	e1
22	1	e2
23	1	e4
24	1	e5
25	1	e6

Questions point here, so "Show in text" for Q1 highlights **e1**.

Gloss and evidence rows are created automatically

Murad never fills these two tables by hand. When he saves, the server reads the passage HTML, finds every `data-g` and `ev` key, and adds or removes rows to match.

passage_gloss_translation

Exercise · translation

The meaning of each marked word, one row per language. This is the main reason BandUp can go global.

gloss_id	language_code	meaning	source	reviewed	who
11 (eliminated)	az	məhv edilmək	HUMAN	true	Murad
11 (eliminated)	en	removed completely	MACHINE	false	drafted (V4)
11 (eliminated)	tr	yok edilmiş	HUMAN	true	Leyla
12 (gov.-sponsored)	az	dövlət tərəfindən dəstəklənən	HUMAN	true	Murad
12 (gov.-sponsored)	en	paid for or supported by the government	MACHINE	false	drafted (V4)
13 (predator)	az	yırtıcı	HUMAN	true	Murad
13 (predator)	tr	yırtıcı	MACHINE	false	auto-translate
14 (elk)	az	maral	HUMAN	true	Murad
14 (elk)	en	a very large kind of deer	MACHINE	false	drafted (V4)

Good to know

- **source**: HUMAN = a person wrote it. MACHINE = a translation tool wrote it.
- **reviewed**: has a person checked it? Leyla's to-do list = all `tr` rows with `reviewed = false`.
- The `en` rows are simple English definitions, the safety net when a language is missing (section 11). All 25 come from V4: drafted by a machine, so they wait as `MACHINE` / `reviewed = false` until a person checks them.

media

Exercise

Images (and later audio). Only information *about* the file; the file itself is in file storage.

id	kind	storage_key	mime_type
5	IMAGE	media/2026/10/3f2a.png	image/png

Also stores `byte_size`, `width` / `height` (images) and `duration_ms` (audio, for Listening).

media_translation

translation

A text description of the image per language, for blind users (screen readers).

media_id	lang	alt_text
5	en	Diagram of a coral reef, labels 1–3
5	az	Mərcan rifinin sxemi, 1–3 nişanlar

9 Group 5: Questions and answers

The Wolves exercise has 3 question groups. This is what the learner sees, then the tables behind it:

```
Questions 1–3   TRUE / FALSE / NOT GIVEN
  1. Wolves disappeared from Yellowstone mainly because of disease.
  2. More wolves were released in the second year than in the first.
  3. The number of tourists increased after the wolves returned.

Questions 4–6   Complete the sentences. Write NO MORE THAN TWO WORDS from the text.
  4. Without wolves, too many elk damaged young trees near ______.
  5. The recovery of plants created homes for beavers and ______.
  6. Critics say it is hard to give credit for the changes only to ______.

Questions 7–8   Choose TWO letters, A–E.
  Which TWO things, according to some ecologists, also reduced elk numbers?
  A disease   B harsh winters   C tourists   D bears   E floods
```

`question_group`

Questions

A block of questions of one type with one instruction, plus the IELTS rules for that block.

id	ex.	pos	type_code	instruction_html	choose_ count	max_ words	numbers_ allowed	options_ reusable	media_id
31	1	1	TFNG	Do the statements agree…	—	—	false	false	—
32	1	2	FILL	Complete the sentences…	—	2	false	false	—
33	1	3	MCMA	Choose TWO letters…	2	—	false	false	—
34	2	1	MATCH_HEADINGS	Choose the correct heading…	—	—	false	false	—
35	2	2	DIAGRAM_LABEL	Label the diagram…	—	2	true	false	5

The heading "Questions 4–6" is **not stored**; it's calculated (section 12). The IELTS-rule columns are explained in section 13.

`question_group_translation`

Questions · translation

An optional translation of the instruction, shown *beside* the official English one to help beginners.

group_id	language_code	instruction_html
31	az	Müddəalar mətndəki məlumatla uyğun gəlirmi? TRUE, FALSE və ya NOT GIVEN seçin.

`question`

Questions

One question. Which text columns are used depends on the type.

id	group	pos	stem_text	stem_pre	stem_post	evidence_id
41	31	1	Wolves disappeared … mainly because of disease.	—	—	21 (e1)
42	31	2	More wolves were released in the second year…	—	—	22 (e2)
43	31	3	The number of tourists increased…	—	—	—
44	32	1	—	Without wolves, too many elk damaged young trees near	.	23 (e4)
45	32	2	—	The recovery of plants created homes for beavers and	.	24 (e5)
46	32	3	—	Critics say it is hard to give credit … only to	.	25 (e6)
47	33	1	Which TWO things … also reduced elk numbers?	—	—	25 (e6)

Good to know

- `stem_text` for statements and questions; `stem_pre` + blank + `stem_post` for gap-fill.
- There is **no question number column**. Q44 is shown as "4" because it's calculated (section 12).
- Q43 (tourists, NOT GIVEN) has no evidence, because the text never mentions tourists.
- The database checks that the evidence belongs to the *same* exercise.

question_option

Questions

The choices a learner picks from. They belong to either **one question** (multiple choice) or **a whole group** (matching), never both.

Owned by question 47 (Choose TWO)

id	group_id	question_id	key	label
91	null	47	A	disease
92	null	47	B	harsh winters
93	null	47	C	tourists
94	null	47	D	bears
95	null	47	E	floods

Owned by group 34 (headings, Coral Reefs)

id	group_id	question_id	key	label
71	34	null	i	Rising sea temperatures
72	34	null	ii	Why reefs matter
73	34	null	iii	Hope from new research
74	34	null	iv	Tourism in Australia

question_answer

Questions

The correct answer(s).

question_id	value	pos
41	FALSE	1
42	TRUE	1
43	NOT GIVEN	1
44	riverbanks	1
44	river banks	2
45	songbirds	1
46	wolves	1
47	B	1
47	D	2

How to read these rows

depends on family

Question	Several rows mean…
Q44 (gap)	**alternatives**: "riverbanks" *or* "river banks" are both correct
Q47 (Choose TWO)	**all required**: B *and* D, 1 point each
Q41 (TFNG)	always exactly one row

question_explanation

Questions · translation

Why the answer is correct, per language. Shown only after the learner submits.

question_id	lang	explanation_html	source	reviewed
41	en	Mainly = largely. The text says hunting, not disease → FALSE.	HUMAN	true
41	az	Mainly = əsasən. Mətndə səbəb ovdur, xəstəlik yox → FALSE.	HUMAN	true
41	tr	Mainly = büyük ölçüde. Metin avlanmayı söylüyor → FALSE.	HUMAN	true
44	en	near = along. You need a place: riverbanks.	HUMAN	true

10 Group 6: Attempts (results)

Aysel opens the Wolves exercise in Azerbaijani, answers, and presses Submit after 3 min 34 s:

```
Q1 FALSE   Q2 TRUE   Q3 FALSE   Q4 river banks   Q5 songbirds   Q6 bears   Q7–8 B, A
```

`attempt`

Attempts

One row per submission: who, which exercise, which version, the time taken and the score.

id (uuid)	user_id	exercise_id	exercise_ revision	support_ language	submitted_at	elapsed_ seconds	score	total
0b91…44	null	1	1	az	2026-10-20 18:02	281	6	8
7f3c…e2	2 (Aysel)	1	2	az	2026-10-22 09:14	214	5	8
c55a…19	3 (Emre)	1	2	tr	2026-10-22 11:40	265	7	8

Good to know

- `user_id = null` means an anonymous visitor (no account). That still works.
- `score` and `total` are **points**. Total is 8 because Q7–8 is worth 2. The percent (5/8 = 63%) is calculated, not stored.
- `support_language` = the language the feedback was shown in.

`attempt_answer`

Attempts

One row per question in an attempt: what was answered, what was correct *at that moment*, and the points.

attempt	question_ id	question_ number	given_ value	expected_ value	points_ awarded	points_ possible	is_ correct
7f3c…	41	1	"FALSE"	["FALSE"]	1	1	true
7f3c…	42	2	"TRUE"	["TRUE"]	1	1	true
7f3c…	43	3	"FALSE"	["NOT GIVEN"]	0	1	false
7f3c…	44	4	"river banks"	["riverbanks","river banks"]	1	1	true
7f3c…	45	5	"songbirds"	["songbirds"]	1	1	true
7f3c…	46	6	"bears"	["wolves"]	0	1	false
7f3c…	47	7	["B","A"]	["B","D"]	1	2	false

Total: 1+1+0+1+1+0+1 = **5 of 8 points**.

- `expected_value` is a **copy** of the answer at grading time, so later edits never rewrite this result.
- `is_correct` is calculated by the database (awarded = possible). Q7–8 got 1 of 2 points, so it is not fully correct.
- `question_number` = the number Aysel saw (7 for "Q7–8").

11 Languages and fallback

The rule: two kinds of text

Kind	Language	Where	Examples
Exam content	English (what is learned)	main tables, written once	passage, questions, options, answers, official instructions
Help text	the learner's language	`*_translation` tables, one row per language	word meanings, explanations, course and stage titles, image descriptions

Button labels ("Submit", "Start again") are neither. They live in translation files in the code, not in the database.

Fallback: Emre opens the Wolves exercise in Turkish

For each word the server asks: **Turkish?** → if missing, **English** (Turkish's `fallback_code`) → if missing, the word isn't marked.

Word	Rows that exist	Emre sees	Why
eliminated (g1)	`az, en, tr`	yok edilmiş	**Turkish exists**
government-sponsored (g2)	`az, en`	paid for or supported by the government	no tr → **English**
predator (g3)	`az, tr (machine)`	yırtıcı	**Turkish exists**
elk (g4)	`az, en`	a very large kind of deer	no tr → **English**

A word is shown unmarked only when it has no meaning in the learner's language *and* no English definition. Since V4, every Wolves word has an English one.

Adding a new language = only new rows

```
UPDATE language SET enabled = true WHERE code = 'ru';                    -- 1. switch Russian on
INSERT INTO passage_gloss_translation VALUES (11, 'ru', 'уничтожены', 'HUMAN', true, now());   -- 2. translate over time
```

No new tables, no new columns, no code change.

12 Question numbers and points

Numbers are **never stored**. The server counts them in order (group position, then question position), and a "Choose TWO" question uses two numbers:

Group	Question id	Points	Learner sees
31 (TFNG)	`41, 42, 43`	1 each	**Q1, Q2, Q3**, heading "Questions 1–3"
32 (FILL)	`44, 45, 46`	1 each	**Q4, Q5, Q6**, heading "Questions 4–6"
33 (MCMA)	47	2	**Q7–8**

> **Why calculate instead of store?**
> If Murad deletes Q2, stored numbers would leave a gap (1, 3, 4…) unless someone renumbers everything. Calculated numbers fix themselves: the questions are simply counted again, 1 to 7.

13 IELTS rules

Column (on question_group)	IELTS instruction	Example
`max_words`	"ONE WORD ONLY" / "NO MORE THAN TWO WORDS"	Group 32: `2`
`numbers_allowed`	"… AND/OR A NUMBER"	Group 35: `true`
`options_reusable`	"You may use any letter more than once"	Headings: `false`
`choose_count`	"Choose TWO letters"	Group 33: `2`
`media_id`	"Label the diagram below"	Group 35 → image 5

Word limit: group 32, max 2 words

Typed for Q4	Result
`riverbanks`	✓ 1 point
`river banks`	✓ 2 words, accepted alternative
`the river banks`	✗ 3 words, over the limit

Choose TWO: Q7–8, correct B + D

Chosen	Points
`B, D`	2 / 2
`B, A`	1 / 2 (Aysel)
`B, A, D`	0 / 2 (too many letters)

The two other types: **SUMMARY_BOX** (fill gaps by choosing words from a box, which works like matching) and **DIAGRAM_LABEL** (label a picture from `media`, which works like gap-fill). All rules are optional, so a relaxed beginner practice can simply leave them empty.

14 Revision vs version

revision: "which edition did the learner answer?"

Goes up when a *published* exercise's content is saved.

Revision 1: Q4 accepts only "riverbanks". An anonymous visitor answers.
Murad adds "river banks" → **revision 2**.
Aysel and Emre answer revision 2.

Each `attempt.exercise_revision` records this, so results stay explainable.

version: "did someone else save first?"

Goes up on *every* save. It protects against two admins overwriting each other.

Murad opens: version 7. A co-author opens: version 7.
Murad saves → version 8 ✓
Co-author saves "version 7" → **409 Conflict**: "Someone else changed this. Reload."

Never shown to learners.

15 Draft, published, archived



Status	Example	Learners see it?	Can be deleted?
`DRAFT`	Coral Reefs (Murad still writing)	no	yes, if it has no attempts
`PUBLISHED`	The Return of the Wolves	yes	no
`ARCHIVED`	an old passage replaced by a new one	no	no; it keeps learners' history

The Wolves exercise has 3 attempts, so the database **refuses** to delete it. Murad archives it instead, and Aysel's history stays.

16 What learners can and cannot see

When Emre opens `/reading/yellowstone-wolves`, the server reads only the safe tables:

✓ Read for learners	🔒 Never read before Submit
`exercise`, `reading_passage`	`question_answer`
`passage_gloss` + translation	`question_explanation`
`question_group` + translation	other users' `attempt` rows
`question`, `question_option`	DRAFT exercises (Coral Reefs → 404)
`media` + translation	`user_credential`, `refresh_token` (never in any response)

```
GET /api/v1/passages/yellowstone-wolves?lang=tr
{
  "id": "yellowstone-wolves", "language": "tr", "direction": "ltr",
  "title": "The Return of the Wolves", "durationSeconds": 300,
  "passageHtml": "… <span class=\"w\" data-g=\"g1\">eliminated</span> …",
  "glosses": { "g1": "yok edilmiş", "g2": "paid for or supported by the government", "g3": "yırtıcı", "g4": "a
very large kind of deer", … },
  "questionGroups": [ { "type": "TFNG", "heading": "Questions 1–3", "questions": [ { "n": 1, "text": "Wolves
disappeared …" } ] },
                     { "type": "MCMA", "chooseCount": 2, "questions": [ { "n": 7, "nTo": 8, "options": [ … ] } ]
} ]
}                                        ← no answers, no explanations anywhere
```

Answers and explanations come back only in the response to **Submit**.

17 Finding exercises: filters

Learners don't follow a locked path. They open a list and filter it. Every filter is optional, and filters combine ("Mini **and** Nature").

Filter	Example	Comes from
Skill	Reading	`exercise.skill`
Format	Mini test / Full passage	`exercise.format`
Level	A2, B1…	`exercise.level`
Question type	True/False/Not Given	`question_group.type_code`
Topic	Nature	`exercise_topic`
Duration	under 10 minutes	`exercise.duration_seconds`
Not done yet	hide what I already did	`attempt` (logged-in users only)

Examples with our 5 exercises

The exercises learners can find (Coral Reefs and Present Perfect are DRAFT, so they never appear)

Exercise	format	level	minutes	question types	topics
The Return of the Wolves	`MINI`	`B1`	5	`TFNG, FILL, MCMA`	Nature
Busy Honeybees	`MINI`	`A2`	4	`TFNG, MCQ`	Nature, Science
Ancient Rome	`FULL`	`C1`	20	`MATCH_HEADINGS, SUMMARY`	History

Learner chooses	Result
Mini tests	Wolves, Honeybees
Mini tests + level A2	Honeybees
True/False/Not Given	Wolves, Honeybees
Topic: History	Ancient Rome
Under 10 minutes	Wolves, Honeybees
Not done yet (Aysel already did Wolves)	Honeybees, Ancient Rome
Full passage + Nature	nothing (Coral Reefs is still a draft)

```
GET /api/v1/exercises?skill=READING&format=MINI&topic=nature&lang=az
{ "items": [
    { "id": "yellowstone-wolves",
      "title": "The Return of the Wolves",
      "format": "MINI", "level": "B1", "durationSeconds": 300,
      "questionCount": 8, "types": ["TFNG","FILL","MCMA"],
      "topics": [ { "slug": "nature", "name": "Təbiət" } ],
      "bestScore": { "score": 5, "total": 8 } },
    { "id": "busy-honeybees",
      "title": "Busy Honeybees",
      "format": "MINI", "level": "A2", "durationSeconds": 240,
      "questionCount": 6, "types": ["TFNG","MCQ"],
      "topics": [ { "slug": "nature", "name": "Təbiət" },
                  { "slug": "science", "name": "Elm" } ] }
  ],
  "page": 0, "size": 20, "totalItems": 2 }
```

Topic names come back in the learner's language (Təbiət). `bestScore` appears only for exercises the logged-in learner has already done. Clicking an item opens it, the learner practises and submits, with nothing locked.

18 Rules the database enforces

The app also checks these, but the database is the last guard: even a bug or a manual SQL command can't break them.

Rule	Rejected example
Timer is optional, but if set: duration > 0 and the warning comes before the end	duration 300, threshold 400 ✗
An option has exactly one owner (a group or a question)	option with both group 34 and question 47 ✗
No duplicate option keys for one owner	two "A" options on Q47 ✗
Evidence must be in the same exercise as the question	Q41 (Wolves) → evidence from Coral Reefs ✗
Evidence still used by a question can't be removed	Murad deletes the e1 sentence while Q1 uses it ✗ (clear error message)
An exercise with attempts can't be deleted	delete Wolves ✗ → archive instead
One translation per item per language	a second `az` meaning for g1 ✗
Languages must exist in `language`	meaning in `'azz'` ✗
Slugs and emails are unique	a second `yellowstone-wolves` ✗
Points awarded ≤ points possible; score ≤ total	3 / 2 points ✗
Word limit and "Choose N" are positive	`max_words = 0` ✗
One password per user; deleting a user deletes their password and tokens	a second credential row for Aysel ✗
Refresh token hashes are unique and exactly 64 lower-case hex characters	`'not-a-hash'` ✗ · an upper-case hash ✗
A refresh token can't expire before it's issued	expires yesterday ✗

19 Future features

Feature	What we add	What stays the same
Listening	`listening_audio` table (uses `media` with kind AUDIO)	exercise, all question tables, attempts, translations
Grammar	maybe nothing; drills are questions with answers	everything
Full mock test (3 passages, 40 questions, band score)	`mock_test` + `band_conversion` tables	exercises are reused as its parts
Vocabulary lists	a shared dictionary table, linked through `passage_gloss.lemma`	glosses keep working
Writing / Speaking	new tables for essays and recordings with criteria scores (teacher or AI)	users, courses, exercises, translations
Google sign-in	a sign-in table next to `user_credential` (e.g. `user_identity`)	`app_user`, roles, refresh tokens

> **The key idea**
>
> Every future feature **adds** tables next to these. None of them requires changing or rebuilding the 28 tables in this guide.